Luyang Zhao

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Research Interest

My current research spans a range of different topics in **Modular robots**, **Aquatic Multi-robot Systems**, **Soft robots**, **Motion Planning**, **Machine Learning**, and **SLAM**.

Education

Dartmouth College — **Hanover, NH**
Ph.D. Candidate in Computer Science — *Sep. 2018–Expected June 2025*
Advisor: Prof. Devin Balkcom; Honors: Neukom Outstanding Graduate Research Prize

University of Minnesota — **Minneapolis, MN**
B.S. in Computer Science — *Graduated **with distinction** in 2018*
Research advisor: Prof. Maria Gini; Honors: Dean's list

B.A. in Mathematics
Senior project advisor: Prof. Karel Prikry; Honors: Dean's list

Research Experience

Design of Soft Modular Robots — Oct. 2020 - Present
Dartmouth College (*Advisor: Prof. Devin Balkcom*) — *Hanover, NH*
- Led the design and development of **Tensegrity-Blocks** and **StarBlocks**, along with various soft modular robotic systems for terrestrial and aquatic use. Focused on designs for autonomous and collaborative tasks, aiming to **integrate modular and soft robotics**. For selected project demonstrations, please refer to the video links provided: Link1, and Link2.

Design of Dolphin Dynamics — Jan. 2024 - Present
Dartmouth College (*Advisor: Prof. Devin Balkcom*) — *Hanover, NH*
- Led the development of **dolphin** dynamics by integrating cable-driven mechanisms and tensegrity structures to replicate marine locomotion.

Design of Aquatic Biomechanics of Locomotion — June 2024 - Present
Georgia Tech (*Advisor: Prof. Daniel Goldman*) — *Atlanta, GA*
- Collaborating on the aquatic robotics project to develop a minimal robophysical model of quadriflagellate self-propulsion, focusing on simplifying the design and improving locomotion efficiency and speed using cable-driven systems.

Aquatic multi-robot Research — Sep. 2018 - Jan. 2019
Dartmouth College (*Advisor: Prof. Alberto Quattrini Li*) — *Hanover, NH*
- Conducted comprehensive research on **water quality assessment** using a fleet of heterogeneous aquatic robotic systems. Developed and implemented advanced trajectory planning algorithms to optimize spatial exploration, enhancing the accuracy and efficiency of water quality data acquisition across extensive aquatic environments.

Motion Planning Research — Sep. 2018 - Nov. 2020
Dartmouth College (*Advisor: Prof. Devin Balkcom*) — *Hanover, NH*
- Conducted research in optimal robot motion strategies, including Bounds Planning and Piecewise Linear Regression Complex. Focused on developing **memory-efficient algorithms** and heuristic approaches for single and multi-robot motion planning.

Undergraduate Research — Sep. 2016 - May 2018
University of Minnesota (*Advisor: Prof. Maria Gini*) — *Minneapolis, MN*
- Developed an **assistive AI device** for individuals with memory loss, integrating voice-activated technologies for cognitive support; Designed an **object-finding robot** utilizing SLAM and advanced object detection algorithms for efficient navigation in unknown environments.

Work Experience

Applied Scientist Intern @ Amazon Robotics — June 2021 – Sep. 2021
Amazon — *Westborough, Massachusetts*
- Conducted in-depth research and development in **localization**, **path planning**, and **navigation** for **Amazon's autonomous mobility robots**, addressing key challenges and enhancing robot performance through the design and implementation of new algorithms integrated using **ROS** and **C++**, significantly improving operational capabilities.

Summer Research Intern @ TuSimple Planning — June 2022 – Sep. 2022
TuSimple — *San Diego, California*
- Developed an advanced **local turn lateral trajectory optimization** for **autonomous trucks** using a **kinematic tractor-trailer model** that enhanced kinematic behavior precision during local turns, and focused on **obstacle avoidance** by generating optimized lateral trajectories through **sequential quadratic programming** to minimize state cost, input cost, and jerkiness within necessary constraints, implementing these solutions in a **real-time environment** with **ROS** and **C++**.

Workshops Organized

Tensegrity Robotics Workshop October 5, 2023
IROS 2023 *Room 250C, Huntington Place, Detroit*

- Co-organized the tensegrity robot workshop with Xiaonan (Sean) Huang, Rebecca Kramer-Bottiglio, Kostas Bekris, Devin Balkcom, Joran Booth, Will Johnson, Kun Wang, and Shiyang Lu, and presented our work "Untethered Self-Reconfiguring Morphable Modular Robots for Field Deployment," which highlighted innovative designs in robotic mobility and adaptability for natural, unstructured environments.

Honors and Awards

Admissions Ambassador for Dartmouth College (2023-2024)

2nd place in the Neukom Outstanding Graduate Research Prize (2023)

Dartmouth's Guarini Travel Award (2023, 2024)

Undergraduate **Research** Opportunities Program (UROP) **Scholarship** (2016)

First prize in Provincial High School **Mathematics** Competition (2011)

Mentorship Experience

Lab Mentor Sep. 2018 – Now
Dartmouth Reality and Robotics Lab *Hanover, NH*

- Master students: Chun-Yi She (2023-now), Yitao Jiang (2022-now), Yijia Wu (2021-2022, now PhD student at WPI), Weishu Zhan (2022)
- Undergraduate students: Josiah Putman (now in Google), Maxine Perroni-Scharf (now PhD student at MIT)

Teaching Assistant Sep. 2018 – Now
Dartmouth College *Hanover, NH*

- **CS89/189**: The Dark Side of AI/ML – 2024 Spring
- **CS81/281** : Principles of Robot Design and Programming – 2018 Fall
- **CS76/276** : Artificial Intelligence – 2018 Winter, 2019 Fall and 2023 Fall
- **CS1** : Introduction to Programming and Computation – 2019 Spring and 2020 Spring
- **CS50** : Software Design and Implementation – 2019 Summer

Teaching Assistant, Summer Computing Academy June 2017
University of Minnesota *Minneaplis, MN*

- Assisted senior high school students in developing programs for Scribbler robots, image processing, video, 3D printing, and other applications.

Manuscripts in Preparation

[1] **Luyang Zhao**, Y.Jiang, M. Chen, D.Balkcom. Versatile Variable-Stiffness Modular Robots: Enhanced Design and Adaptive Control for Multifunctional Performance in Varied Aquatic Environments. (In preparation to submit to RA-L in July)

[2] **Luyang Zhao**, Y. Jiang, C. She, H. Dong, M. Chen, D.Balkcom. Designing Dolphin Dynamics: Integrating Cable-Driven Mechanisms and Tensegrity Structures for Replicating Marine Locomotion" (In preparation to submit to RA-L in August)

[3] M. Jeong, A. Chadda, Z. Ren, **Luyang Zhao**, H. Liu, M. Roznere, A. Zhang, Y. Jiang, S. Achong, S. Lensgraf, and A. Q. Li. SeeSea: Multi-modal 3D Perception Dataset of In-water Obstacles for Navigation of Autonomous Surface Vehicles" (In preparation to submit to IJRR in Sep.)

[4] Y. Jiang, **Luyang Zhao**, M. Chen, D. Balkcom. Design and Implementation of an Untethered Underwater Glider Using Syringe-Based Actuation Mechanisms" (In preparation to submit to ICRA in Sep.)

Manuscripts Under Review

[5] **Luyang Zhao**, Y. Jiang, K. Bekris, and D. Balkcom. Tensegrity-Blocks: Modular Shape-changing Blocks Enable Self-assembling Robotic Structures. ***Science Robotics.*** (Under revision)

[6] **Luyang Zhao**, Y. Jiang, C. She, D.Balkcom, H. Dong, M. Chen. Design and Experiment of a Lightweight Robotic Tensegrity Morphing Wing. *AIAA SciTech2025* (Under review)

Publications

[7] **Luyang Zhao**, Y. Wu, W. Yan, W. Zhan, X. Huang, J. Booth, A. Mehta, K. Bekris, R. Kramer-Bottiglio, and D. Balkcom. Starblocks: Soft actuated self-connecting blocks for building deformable lattice structures. *IEEE Robotics and Automation Letters*, 8(8):4521–4528, 2023

[8] **Luyang Zhao**, Y. Wu, J. Blanchet, M. Perroni-Scharf, X. Huang, J. Booth, R. Kramer-Bottiglio, and D. Balkcom. Soft lattice modules that behave independently and collectively. *IEEE Robotics and Automation Letters*, 7(3):5942–5949, 2022

[9] **Luyang Zhao**, J. Putman, W. Wang, and D. J. Balkcom. PLRC*: a piecewise linear regression complex for approximating optimal robot motion. In *International Conference on Intelligent Robots and Systems, IROS 2020*

[10] J. Putman, L. Oh, **Luyang Zhao**, E. Honnold, G. Brown, W. Wang, and D. J. Balkcom. Piecewise linear regressions for approximating distance metrics. *CoRR*, abs/2002.12466, 2020

[11] J. Putman, L. Oh, **Luyang Zhao**, E. Honnold, G. Brown, W. Wang, and D. J. Balkcom. LLDM: locally linear distance maps for robot motion planning: Extended abstract. In *2019 International Symposium on Multi-Robot and Multi-Agent Systems, MRS 2019*

[12] L. Ferland, Z. Li, S. Sukhani, J. Zheng, **Luyang Zhao**, and M. L. Gini. Assistive AI for coping with memory loss. In *The Workshops of the The Thirty-Second AAAI Conference on Artificial Intelligence, 2018*, AAAI Workshops

[13] M. Jeong, A. Chadda, Z. Ren, **Luyang Zhao**, H. Liu, M. Roznere, A. Zhang, Y. Jiang, S. Achong, S. Lensgraf, and A. Q. Li. Multi-modal perception dataset of in-water objects for autonomous surface vehicles. In *IEEE ICRA Workshop on Field Robotics 2024*, 2024

Conference and Seminar Presentations

[1] "Self-assembling Soft Modular Robots for Manipulation," selected talk at the New England Manipulation Symposium (NEMS) 2024, 9:45-10:00.
[2] "StarBlocks: Soft Actuated Self-Connecting Blocks for Building Deformable Lattice Structures." Presented at RoboSoft 2024.
[3] "Untethered Self-Reconfiguring Morphable Modular Robots for Field Deployment" Presented at ICRA 2023 Tensegrity workshop
[4] "StarBlocks: Soft Actuated Self-Connecting Blocks for Building Deformable Lattice Structures." Initially presented part of the idea at the ICRA 2022 MSRR Workshop, and at NERC 2022.
[5] "Soft Modular Robotics: Design and Applications" invited talk at Rutgers university robotics seminar (10/17/2022 10:30am-12:00am)
[6] "Soft Lattice Modules That Behave Independently and Collectively," Presented at Robosoft 2022
[7] "PLRC*: a piecewise linear regression complex for approximationg optimal robot motion", Presented at IROS 2020
[8] "LLDM: locally linear distance maps for robot motion planning", Presented at MRS 2019

Review Activities

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
IEEE-RAS International Conference on Soft Robotics (RoboSoft)
IEEE RAS EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob)

References

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Prof. Muhao Chen
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University of Kentucky — Email: muhaochen@uky.edu

Prof. Xiaonan (Sean) Huang
Assistant Professor, Robotics
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Prof. Rebecca Kramer-Bottiglio
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